

# Evanns Morales-Cuadrado

U.S. Citizen

## Robotics PhD Candidate Georgia Institute of Technology

### CONTACT

Email: [evannsmc@gmail.com](mailto:evannsmc@gmail.com)

Github: <https://github.com/evannsmc>

Research Website: <https://www.evannsmc.com/>

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### TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, C#/.NET, Matlab/Simulink
- **Robotics & Control Frameworks:** ROS 2, NATS, PX4, ArduPilot, acados, CasADi
- **Machine Learning & Simulation:** JAX, PyTorch, MuJoCo, Reinforcement Learning
- **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
- **Hardware & Experimentation:** UAV assembly, repair, and deployment
- **Motion-Capture:** OptiTrack and Vicon
- **Languages:** Spanish (Native), English (Native)

### EDUCATION

**Georgia Institute of Technology** | 2022-2027

Ph.D. in Robotics

*Doctoral Minor in Optimization Methods*

- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

**University of Texas at Arlington** | 2018-2022

Honors Bachelor of Science in Electrical Engineering

*Minor in Mathematics*

*Minor in Physics*

*Certificate in Unmanned Vehicle Systems*

- GPA: 3.9/4.0 Summa Cum laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

### WORK EXPERIENCE

**Graduate Researcher** | AI for Autonomy

May 2026 - Present

Sandia National Laboratories

Albuquerque, NM

- Conducted applied research using reinforcement learning for autonomous systems
- Developed C#/.NET networking frameworks and execution scripts for coordinating communication and task execution across multi-robot systems
- Created novel robotics and automation infrastructure to advance national interests

**Graduate Research Assistant** | FACTS Lab

August 2022 - Present

Georgia Institute of Technology

Atlanta, GA

- Advised by Dr. Samuel Coogan
- Implemented novel aggressive tracking control methods on quadrotor hardware
- Developed novel nonlinear control methods for safe autonomy
- Published novel research at top conferences and journals in the fields of robotics and control theory

**Research Assistant** | Autonomous Systems Lab

November 2021 - May 2022

University of Texas at Arlington

Arlington, TX

- Advised by Dr. Frank Lewis
- Received personal mentorship from a leading researcher in the field of controls and reinforcement learning (ranked 12<sup>th</sup> in the USA and 23<sup>rd</sup> in the world during my time in his lab)
- Aided graduate students on reinforcement learning research applied to quadrotor control

**Research Assistant** | Dynamical Networks and Control Lab  
University of Texas at Arlington

October 2019 - May 2022  
Arlington, TX

- Advised by Dr. Yan Wan
- Gained experience with Robot Operating System (ROS), OpenCV, and path planning
- Original research in learning-based minimum time and energy path-planning for multi-vehicle systems

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**SELECTED  
PUBLICATIONS**

**E. Morales-Cuadrado**, L. Chung, S. Kousik and S. Coogan, “RTD-RAX: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances.” Joint Submission to 2026 MECC-ALDSC. (*under revision*) [preprint available](#).

L. Baird, **E. Morales-Cuadrado**, and S. Coogan, “Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic.” Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. (*under revision*) [preprint available](#).

**E. Morales-Cuadrado**, C. Llanes, Y. Wardi and S. Coogan, “Newton-Raphson Flow for Aggressive Quadrotor Tracking Control.” [2024 American Control Conference \(ACC\)](#)

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**OPEN-SOURCE  
CONTRIBUTIONS**

- **PX4 Autopilot**: Contributed documentation improvements to the official PX4 Autopilot manual.
- **Vicon4PX4**: Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech’s Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.

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**SERVICE &  
LEADERSHIP**

**Co-Founder and President of Puerto Rican Student Association**  
Georgia Institute of Technology

August 2024 - Present  
Atlanta, GA

- Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture
- Formed a three-member co-founder board and identified a faculty advisor
- Raised funds for the organization and recruited members from campus
- Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech

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**HONORS AND  
AWARDS**

**Top-3 Finalist at a Deep Learning Research Symposium**

November 2023

Professor-sponsored award for novel research in deep learning and robotics

**Goizueta Fellowship at Georgia Tech**

August 2022

Highly selective fellowship for graduate students of Latin-American descent

**Summa Cum Laude Honors at University of Texas at Arlington**

May 2022

Exceptional academic distinction in top GPA tier for graduation honors

**Electrical Engineering Honors Scholar at University of Texas at Arlington**

December 2021

Inaugural member of the electrical engineering honors cohort

**IEEE-HKN International Electrical Engineering Honor Society Membership**

August 2021

Selective undergraduate society recognizing outstanding academic achievement and community service

**Chance Vought Engineering and Science Endowment Scholarship**

August 2020

Highly selective award given annually to three recipients across the entire College of Engineering

**National Hispanic Scholar**

March 2018

Recognition of top national performers of Latin descent on the SAT

**National Merit Scholar**

March 2018

Recognition of top national performers on the SAT